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TY ECE/AIML

PBL 4- Project Synopsis

Title of the Project Edge-AI Based Automated Hematology Screening System Using Digital Microscopy

Domain of Project:

- Artificial Intelligence and Machine Learning
- VLSI and Embedded Systems
- Applied Electronics
- Healthcare Informatics

Introduction / Background

- Peripheral blood smear (PBS) examination is the gold standard for diagnosing malaria, anemia, and leukemia; however, the traditional workflow is heavily reliant on manual microscopy. This process is time-consuming (10–20 mins per slide) and subject to observer fatigue and variability, leading to diagnostic delays—particularly in rural India where qualified hematologists are scarce.
- While high-end automated digital pathology systems exist, their high cost (₹10–15 lakhs) and infrastructure demands make them inaccessible to small-scale diagnostic centers. There is an urgent need for a "middle-ground" solution that leverages recent advances in **edge computing** (Raspberry Pi/Jetson Nano) and **lightweight CNNs** (YOLOv8-Nano/MobileNet) to provide localized, low-cost, and automated screening.
- This project addresses the pathologist shortage by developing an AI-assisted triage tool that is both socially impactful and technically rigorous. The study is structured across three phases - moving from initial AI inference to full hardware automation and field validation - aligning with the ECE/AIML curriculum to produce a research-oriented, patentable diagnostic platform.

Problem Statement

- The objective of this project is to develop a **low-cost, edge-AI-based automated haematology screening system** that retrofits onto standard laboratory microscopes. The system will automate digital image acquisition and perform on-device analysis using lightweight deep learning models (such as YOLOv8-Nano) to detect haematological abnormalities without requiring constant internet connectivity.

- By creating a low-cost, retrofittable automation system, we can:

    1. **Bridge the Rural-Urban Gap:** Enable high-quality hematological screening in areas lacking resident pathologists.

    2. **Standardise Diagnostics:** Ensure consistent accuracy through AI-driven inference that does not suffer from human fatigue.

    3. **Enhance Scalability:** Provide a tool that allows a single pathologist to review flagged, image-backed reports from multiple remote centres simultaneously.

- **Phase 1 (PBL-4) Focus:** To establish a functional prototype capable of digital image acquisition, executing AI inference on a Raspberry Pi 4, and generating structured diagnostic reports to assist medical professionals in rapid decision-making.

Objectives of the Project

Primary Objectives (PBL-4 Implementation)

1. **Development of a Digital Imaging Interface:** To design and fabricate a custom 3D-printed adapter that integrates a Raspberry Pi 4 and a high-resolution camera module with a standard optical microscope, ensuring stable and consistent digital image acquisition from blood smear slides.
2. **Deployment of Lightweight AI Models:** To train and optimize lightweight deep learning architectures (such as YOLOv8-Nano or MobileNetV2) for the on-device detection and classification of hematological abnormalities, specifically focusing on malaria parasites and sickle cell morphology.
3. **Implementation of an Edge-Processing Pipeline:** To build a localized software framework for real-time image preprocessing (including color normalization and contrast enhancement) and inference, achieving a balance between diagnostic accuracy and acceptable latency on embedded hardware.
4. **Automated Diagnostic Reporting:** To develop a module that generates structured, human-readable PDF reports containing AI-annotated regions of interest (ROIs) and probability scores, facilitating rapid verification by medical professionals.

Future Objectives (Capstone Foundation)

5. **Modular System Architecture:** To design the software and hardware interfaces to be modular, allowing for the seamless integration of X-Y-Z stage automation and autofocus mechanisms in subsequent Capstone phases.
6. **Cloud-Integrated Data Management:** To establish a preliminary secure data pipeline (utilizing Firebase or similar) for the storage of de-identified clinical images, laying the groundwork for future federated learning and remote consultation.
7. **Performance Benchmarking:** To conduct a comparative analysis between the proposed edge-AI system and traditional cloud-based solutions, evaluating metrics such as F1-score, power consumption, and total hardware cost-effectiveness.
8. **Academic Publication:** To document the system's architecture and experimental findings for submission to a Scopus-indexed conference, establishing the research's novelty in the field of low-cost medical electronics.

Scope of the Project

❖ In-Scope for PBL-4 (Phase 1)

The current phase focuses on the fundamental hardware-software integration and the validation of the AI inference engine:

- **Microscope Retrofitting:** Design and 3D-printing of a custom adapter to mount the Raspberry Pi HQ Camera onto a standard optical microscope for digital image acquisition.
- **Image Processing Pipeline:** Development of scripts for high-resolution capture and preprocessing, including color normalization, noise reduction, and image tiling.
- **Edge-AI Deployment:** Training lightweight CNNs (YOLOv8/MobileNet) on public datasets (e.g., NIH Malaria Dataset) using Google Colab, followed by optimization and deployment via TensorFlow Lite for on-device inference on Raspberry Pi 4.
- **User Interface & Reporting:** Creation of a lightweight dashboard (using Streamlit or Flask) to visualize real-time inference and generate structured PDF reports featuring annotated regions of interest.

❖ In-Scope for Capstone (Phase 2)

The following features are explicitly reserved for the Capstone project:

- **Mechanical Automation:** Full-scale X-Y-Z axis stage automation using stepper motors and the implementation of auto-focusing algorithms.
- **Clinical Validation:** Testing on live patient samples and routine clinical screening, which requires institutional ethics committee (IEC) approval.
- **Advanced Disease Coverage:** Extension of the model to detect complex conditions like leukaemia subtypes or rare blood parasites beyond the initial malaria and sickle cell focus.
- **Regulatory Certification:** Navigating CDSCO/ICMR certification processes for medical device equivalence.

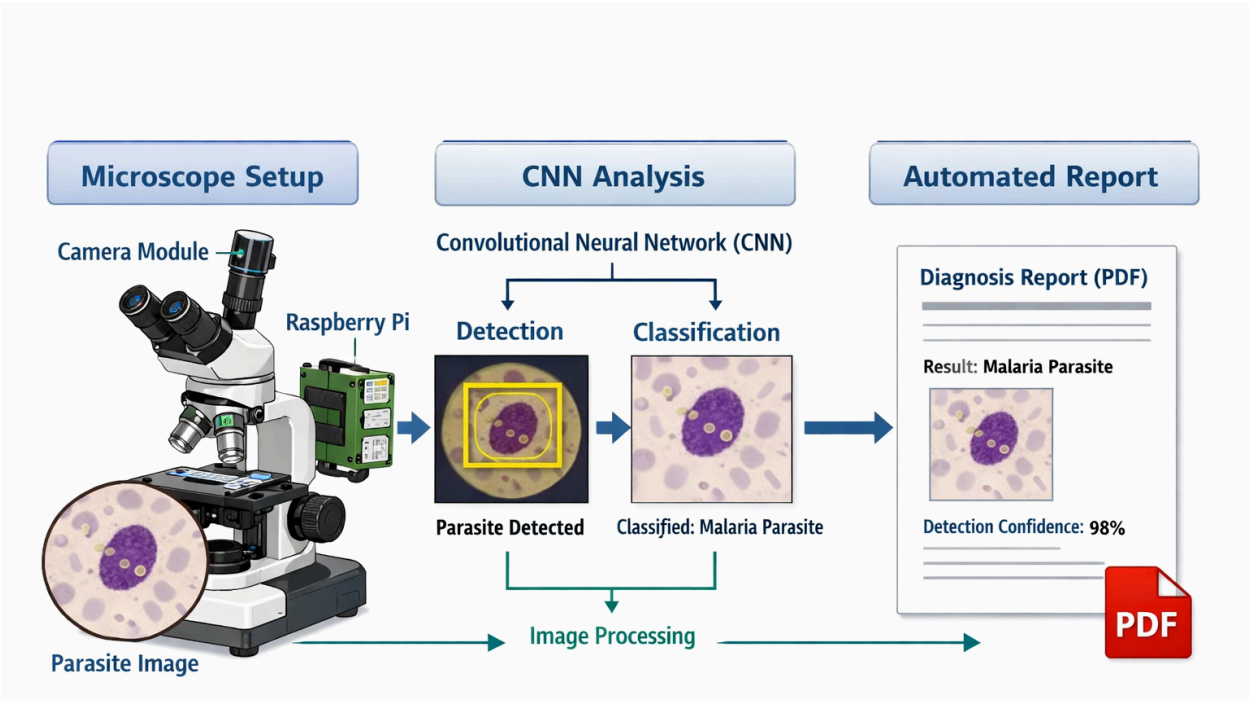
Boundaries and Assumptions

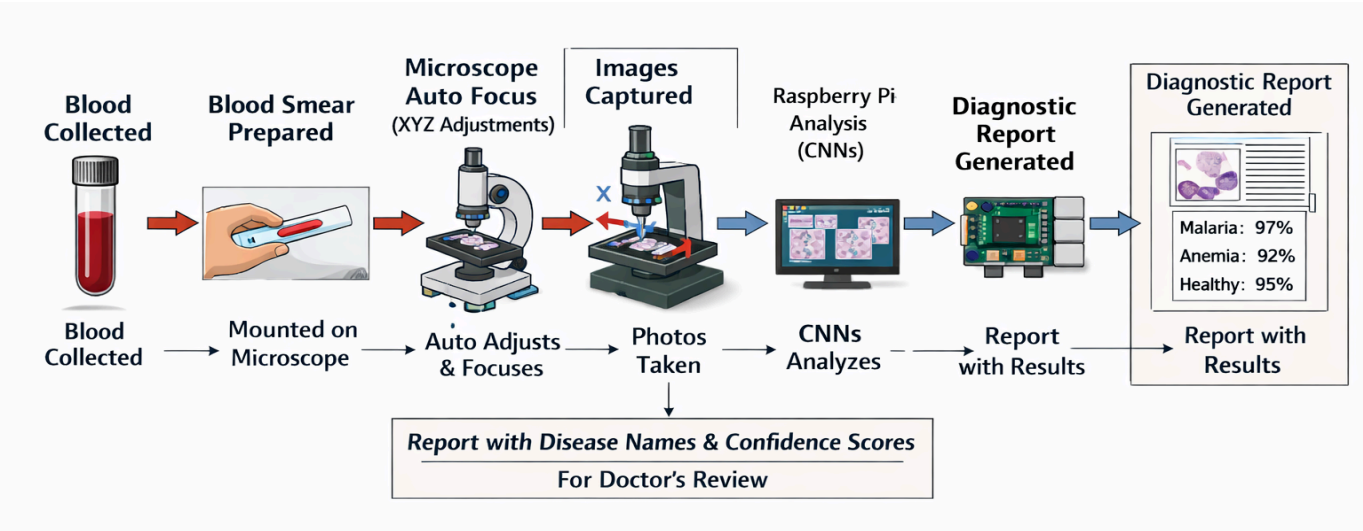
The project operates within the following constraints:

- **Decision-Support Tool:** The system is designed as a **triage/decision-support tool** to assist pathologists, not as a standalone diagnostic device. The final clinical responsibility remains with the medical professional.
- **Technical Feasibility Focus:** Phase 1 (PBL-4) prioritises proving technical feasibility—ensuring image quality and inference latency are acceptable—rather than achieving final product commercialisation.
- **Data Reliability:** It is assumed that publicly available datasets are sufficiently representative for initial training; however, the potential for "domain shift" when moving to local clinical data is acknowledged as a challenge for future phases.

Methodology / Approach

- Block diagram/Flow chart of the project.





- Hardware Platform:
- Software Platform:

  Google Colab

  Kaggle

  Medium

  Coursera

  GitHub
- Data collection and analysis.

  Kaggle

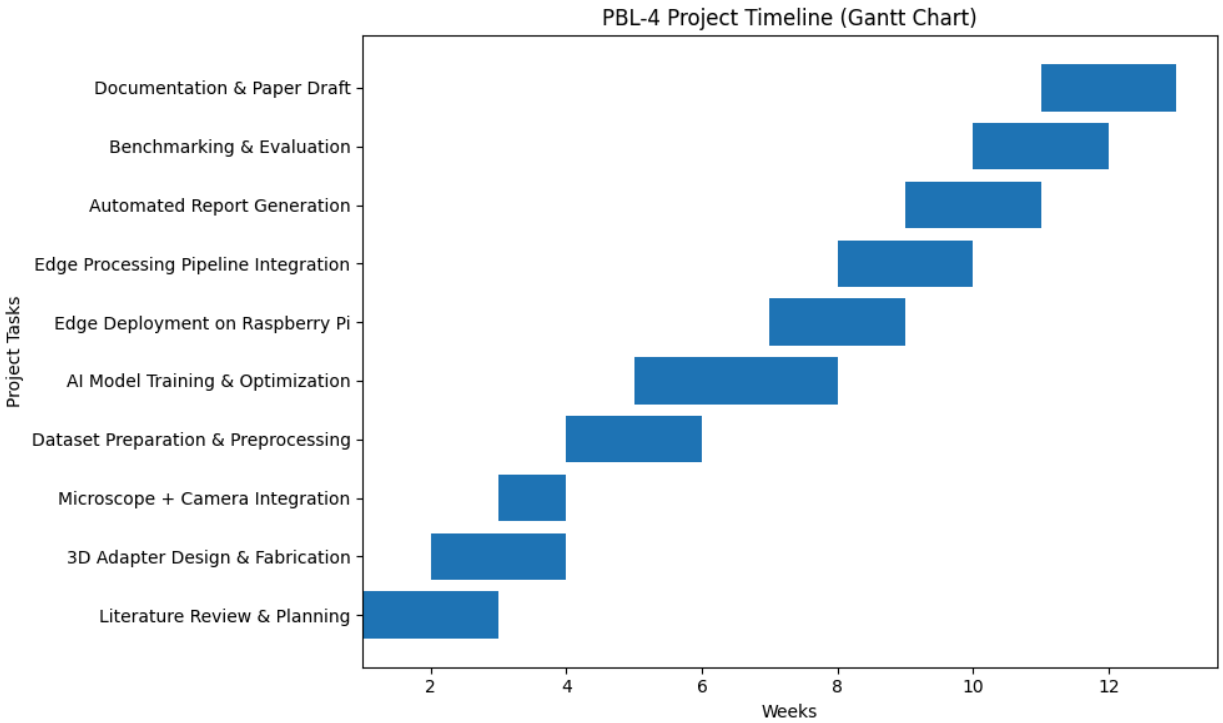
  Mendely Data

  Science Direct

  Broad BioImage Benchmark Collection

Project Timeline

- Gantt chart showing major milestones.



Resources Required (Mention any special support, data, infrastructure, or access.)

Hardware Platform

Microscope: Standard laboratory optical microscope

Compute Unit: Raspberry Pi 4 Model B

Camera: Raspberry Pi HQ Camera with microscope adapter

Stepper Motors: Precision stepper motors for XYZ stage movement (Capstone phase)

Motor Drivers: A4988 / DRV8825 / TMC2209 (Capstone phase)

Linear Motion System: Lead-screw or linear slider mechanism (Capstone phase

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Details of the project group members:

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